MIL-M-60183(MU)
18 March 1965
SUPERSEDING
FA(FC)-PD-87
29 April 1961

MILITARY SPECIFICATION

MAGNETIC DISK, DATA STORAGE: 10526600

1. SCOPE

1.1 This specification covers one type of magnetic disk, hereinafter referred to as the memory, part of the Gun Direction Computer: M18. The magnetic disk is used to store information.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-N-6011	Nitrogen Gas
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18; Manufacturing Processes and Inspection of
MIL-M-60185	Magnetic Disk, Data Storage: Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

FSC 1220

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DRAWINGS

U.S. Army Munitions Command

F10526600	Magnetic Disk, Data Storage
D10526509	Housing Assembly
F10526512	Rear Headplate, Wiring Diagram
F10526513	Front Headplate, Wiring Diagram

PACKAGING DATA SHEET

MIL-P-14232/P10526600	Packaging of Magnetic Disk, Data Storage

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Design and construction.- The memory shall be designed and constructed in accordance with Drawing F10526600.

3.2.1 Disk rotation.- The memory disk shall turn in a counter clockwise direction as viewed from the side of the memory opposite the motor when the motor power is applied.

3.2.2 Residue.- Immediately prior to the sealing and gassing requirements of 3.2.3 and 3.2.4, the memory shall perform an operational run of 150 starts, up to full speed, and 150 complete stops without any evidence of iron oxide residue being deposited on the head plates.

3.2.3 Sealing.- Subsequent to meeting the requirements of 3.2.2 the equipment shall be pressurized to 27 ± 2 inches of mercury, gage and held at this pressure for a period of 5 minutes without pumping. The pressure drop during the holding period shall not exceed 0.1 inch of mercury.

3.2.4 Gassing.- The memory shall be flushed with nitrogen gas, MIL-N-6011, Grade A, Type 1, to a pressure of 32 ± 2 inches of mercury after evacuating to 10,000 microns. Upon completion of flushing, the unit shall be filled with nitrogen gas, previously specified, to a pressure of 32 ± 2 inches of mercury and the unit shall then be sealed.

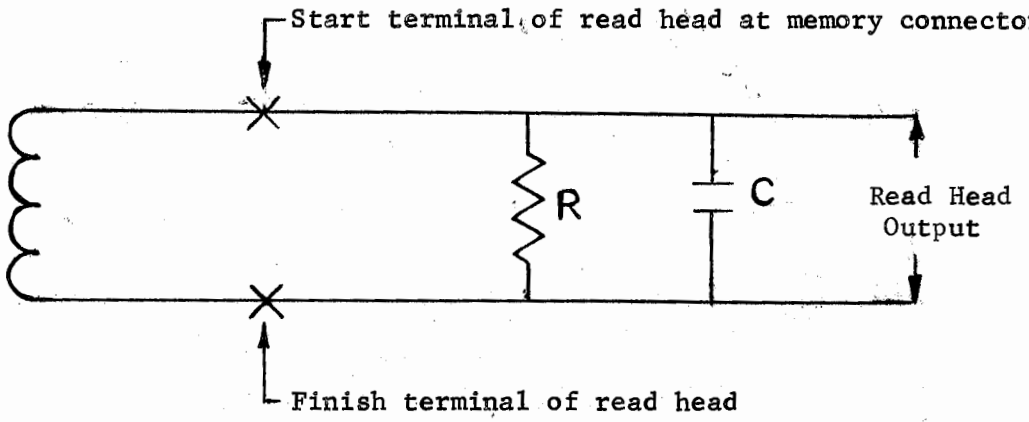
3.2.5 Starting torque.- The memory disk shall achieve a speed greater than 5000 revolutions-per-minute (rpm) in a period not greater than 1 minute after the motor power has been applied, under the conditions of Table I and Method I. The specified speed shall be achieved within 20 seconds under Method II condition (see 6.3.7).

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TABLE I

	Read Head Load See Figure 1 and 2		Voltage - Line to Neutral (RMS)	Frequency CPS	Temperature in Degrees Fahrenheit
	R (ohms)	C (pf)			
Min.	6.2K	68	100	380	-40
Max.	8.1K	86	140	420	+135

3.2.6 Speed versus voltage variation.- The memory motor speed variation shall not be greater than 6 percent of the high voltage speed when the 3 phase 400 cycle per second ± 0.5 percent voltage, is varied from 100 to 140 volts (v) ± 1.5 percent RMS line to neutral after 1 hour or more of motor operation.



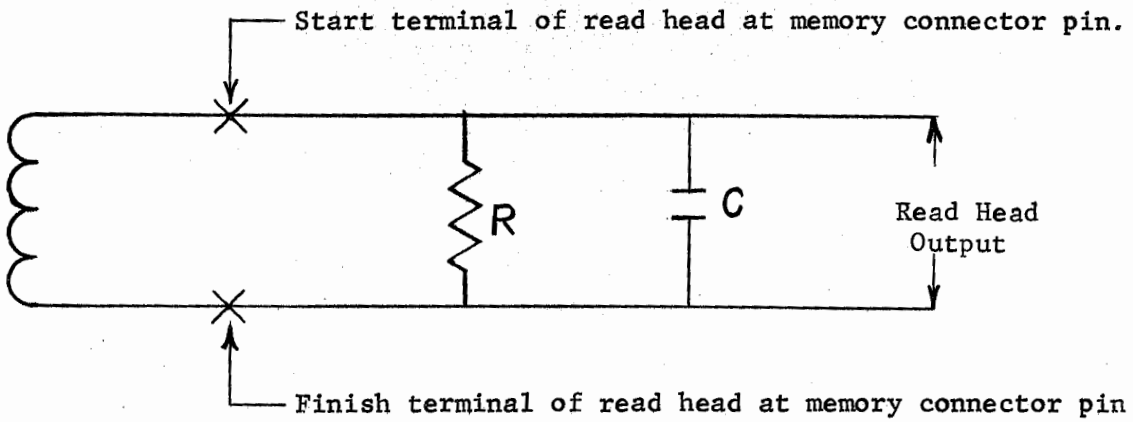
R shall be 6.2K ohms ± .5 percent including input impedance of measuring device.

C shall be 68 pico farads ± .5 percent. This shall include the total load capacitance on the read head external to the memory.

FIGURE 1

Read Head Load A

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R shall be 8.1K ohms ± .5 percent including the input impedance of the measuring device.

C shall be 86 pico farads ± .5 percent. This shall include the total load capacitance on the read head external to the memory.

FIGURE 2

Read Head Load B

3.3 General specifications.- The memory shall meet the following requirements of MIL-M-60185, MIL-C-46357, and MIL-F-13926. Should any conflict occur between these general specifications, the order of precedence shall be MIL-M-60185, MIL-C-46357, and MIL-F-13926.

3.3.1 MIL-M-60185

All requirements

3.3.2 MIL-C-46357

(a) Wiring
(b) Threaded fasteners

3.3.3 MIL-F-13926

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Workmanship

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4

3.4 Environmental.

3.4.1 Vibration.- The memory, mounted in its normal operational position (with its mounting pads), shall withstand vibrations for 10 minutes in each of its three mutually perpendicular axes at frequencies between 0 and 400 cycles per second (cps). Between 0 and 10 cps, the amplitude shall be 0.030 inch (0.060 inch total excursion). Between 11 and 20 cps, the amplitude shall be 0.015 inch (0.030 inch total excursion). Between 21 and 400 cps, the amplitude shall be constrained to an acceleration of 4.5 gravitational units (g's). Total time of vibrations shall be 30 minutes. Subsequent to vibrations, the memory shall meet the requirements of 3.2.5, 3.2.6, and 3.5.

3.4.2 Shock.- The memory shall be capable of withstanding 18 shock impulses in the three mutually perpendicular axes. The acceleration of the shock impulses shall be 5 g's. Pulse duration of each shock impulse shall be 11 ± 1 milliseconds. Subsequent to shock, the memory shall not have been damaged or adversely affected and shall meet the requirements of 3.2.5, 3.2.6 and 3.5.

3.4.3 Temperatures.

3.4.3.1 Operating.- The memory shall meet the requirements of 3.2.5, 3.2.6 and 3.5 while exposed to temperatures of -40°F and +135°F.

3.4.3.2 Storage.- The memory shall meet the requirements of 3.2.5, 3.2.6 and 3.5 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.4.4 Humidity.

3.4.4.1 Operating.- The memory shall meet the requirements of 3.2.5, 3.2.6 and 3.5 while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.4.4.2 Non-operating.- The memory shall meet the requirements of 3.2.5, 3.2.6 and 3.5 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.4.5 Fungus resistance.- The memory shall show no evidence of damage and shall meet the requirements of 3.2.5, 3.2.6 and 3.5 after being inoculated with the composite spore suspension specified in MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.4.6 Elevation.

3.4.6.1 Operating.- The memory shall meet the requirements of 3.2.5, 3.2.6 and 3.5 at an altitude of 10,000 feet above sea level at a temperature of 75 ± 15°F.

3.4.6.2 Non-operating.- The memory shall meet the requirements of 3.2.5, 3.2.6 and 3.5 after exposure to an altitude of 50,000 feet above sea level at a temperature of -65°F for a period of 8 ± 1 hours.

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3.5 Performance.- Unless otherwise specified, the memory shall meet the performance requirements of this specification at ambient temperatures between 60° and 90°F; a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%.

NOTE: Where an operation requirement is to be met, and when applicable, there shall be sufficient air circulation to maintain the memory temperature below a maximum of 165°F.

3.5.1 Memory ground continuity.- There shall be electrical continuity between Pin 1 of the motor power plug, see drawing D10526509, Housing Assembly, and the front and rear headplates. There shall be electrical continuity between pin 1 of the motor power plug and the front and rear headplate connector screws. Electrical continuity shall mean that the direct current resistance is not greater than 0.4 ohms.

3.5.2 Clock channel signal.

3.5.2.1 Clock cycle count.- The clock channel shall contain 4608 negative to positive potential transitions per revolution.

3.5.2.2 Signal amplitude.- The clock channel shall provide a zero-to-peak signal, approximately sinusoidal, which is not less than 42.5 millivolts (mv) when operating under conditions of Table I.

3.5.2.3 Signal frequency (spacing) modulation.- That portion of the clock channel signal form at the leading edge of the positive potential pulse, ± 30 percent from zero potential shall have 576 negative to positive potential changes when any disk sector of 0.785398 ±.000035 radians passes under the clock read head and 288 when any disk sector of 0.392699 ± .000035 radians passes under the clock read head.

3.5.3 Sector channel.

3.5.3.1 Signal amplitude.- All pulses from the sector read head shall not be less than 65mv for Method I, (see 6.3.7 Methods) or less than 70mv for Method II zero to peak, and not greater than 150mv for Method I, or greater than 145mv for Method II zero to peak, when operating under conditions of Table I.

3.5.3.2 Information.- The sector channel shall contain the information as listed in Table II. The (ones or 1's) and the (zeros or 0's) are defined in 6.3.

3.5.3.3 Noise.- There shall be no pulses greater than 10mv, zero to peak, of opposite potential to that which was read out, when reading the information recorded per Table II, other than those pulses required per Table II, when operating under conditions of Table I.

3.5.3.4 Information phase shift.- The sector information, at the 37.5mv crossings of a negative-to-positive potential change, shall be .5 ± .08 clock cycles phase shift with respect to the zero current crossing at the sector write head when writing.

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3.5.3.5 Information pulse width.- The sector information pulse width shall be 1 ± 0.12 clock cycles at 37.5 ± 2 mv above zero potential when operating under conditions of Table I.

3.5.4 Other channels.

3.5.4.1 Channel signal amplitude.- The channel signal amplitude shall be the same as that specified for the sector channel amplitude except that the write head shall be supplied with a write current, which has a steady state amplitude of 65 ± 5 milliamps (ma) and a rise and fall time of 0.09 clock cycles between 10 and 90 percent of the steady state write current. The write current shall not exceed 110 percent of the steady state current and shall not exceed 102 percent of the steady state current for more than .09 clock cycles.

3.5.4.2 Pulse width.- The pulse width shall be 1 ± .12 clock cycles at a channel signal level which is 37.5 ± 2mv above the base line.

3.5.4.3 Write head to read head spacing.- The period between initiating the writing of the flux transition (that point at which the algebraic difference of the write current through the start terminal to center tap and final terminal to center tap is zero) and the 25mv level on the leading edge of the first pulse when reading a one (1) in a field of zeros (0) at the channel read head, shall be equivalent to the period required for the number of clock cycles (designated in Table III) to be read at the clock read head. These requirements to be met under conditions of Table I.

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Table II

Sector	Tp	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Txs	Tx
										Sector							T.T.							F.D.		Sector												
000	0	0	0	0	0	1	1	0	0	0	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	0	0	0	0	0	0	1	1	1	1	1
001	0	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	0	0	0	0	0	1	0	1	1	1	1
002	0	0	0	0	0	1	1	0	0	0	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	1	1	1
003	0	0	0	0	0	1	1	0	0	0	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	0	0	0	0	1	0	0	1	1	1	1
004	0	0	0	0	0	1	1	0	0	0	0	1	0	1	1	1	1	1	0	0	0	1	0	1	0	0	0	0	0	1	0	1	1	1	1	1
005	0	0	0	0	0	1	1	0	0	0	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1	0	0	0	0	1	1	0	1	1	1	1
006	0	0	0	0	0	1	1	0	0	0	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0	0	0	0	0	1	1	1	1	1	1	1
007	0	0	0	0	0	1	1	0	0	0	1	0	0	0	1	0	0	1	1	0	0	0	1	1	1	0	0	0	1	0	0	0	1	1	1	1
010	0	0	0	0	0	1	1	0	0	0	1	0	0	1	1	1	1	0	0	0	0	0	1	1	0	0	0	0	1	0	0	1	1	1	1	1
011	0	0	0	0	0	1	1	0	0	0	1	0	1	0	0	1	1	1	0	0	1	1	1	0	1	0	0	0	1	0	1	0	1	1	1	1
012	0	0	0	0	0	1	1	0	0	0	1	0	1	1	0	0	1	1	1	0	1	1	1	1	0	0	0	0	1	0	1	1	1	1	1	1
013	0	0	0	0	0	1	1	0	0	0	1	1	0	0	1	1	0	0	0	0	0	0	1	0	0	0	0	0	1	1	0	0	1	1	1	1
014	0	0	0	0	0	1	1	0	0	0	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	1	1	1	1	1
015	0	0	0	0	0	1	1	0	0	0	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	0	0	0	1	1	1	0	1	1	1	1
016	0	0	0	0	0	1	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	1	1	1	1	1	1	1	1
017	0	0	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	0	0	1	0	0	0	0	1	1	1	1
020	0	0	0	0	0	1	1	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	0	0	1	0	0	0	1	1	1	1	1
021	0	0	0	0	0	1	1	0	0	1	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	0	0	1	0	0	1	0	1	1	1	1
022	0	0	0	0	0	1	1	0	0	1	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	0	0	1	0	0	1	1	1	1	1	1
023	0	0	0	0	0	1	1	0	0	1	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	0	0	1	0	1	0	0	1	1	1	1
024	0	0	0	0	0	1	1	0	0	1	0	1	0	1	1	1	1	1	0	0	0	1	0	1	0	0	0	1	0	1	0	1	1	1	1	1
025	0	0	0	0	0	1	1	0	0	1	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1	0	0	1	0	1	1	0	1	1	1	1
026	0	0	0	0	0	1	1	0	0	1	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0	0	0	1	0	1	1	1	1	1	1	1
027	0	0	0	0	0	1	1	0	0	1	1	0	0	0	1	0	0	1	1	0	0	0	1	1	1	0	0	1	1	0	0	0	1	1	1	1
030	0	0	0	0	0	1	1	0	0	1	1	0	0	1	1	1	1	0	0	0	0	0	1	1	0	0	0	1	1	0	0	1	1	1	1	1
031	0	0	0	0	0	1	1	0	0	1	1	0	1	0	0	1	1	1	0	0	1	1	1	0	1	0	0	1	1	0	1	0	1	1	1	1
032	0	0	0	0	0	1	1	0	0	1	1	0	1	1	0	0	1	1	1	0	1	1	1	1	0	0	0	1	1	0	1	1	1	1	1	1
033	0	0	0	0	0	1	1	0	0	1	1	1	0	0	1	1	0	0	0	0	0	0	0	1	0	0	0	1	1	1	0	0	1	1	1	1
034	0	0	0	0	0	1	1	0	0	1	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	1	1	1	1	1
035	0	0	0	0	0	1	1	0	0	1	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	0	0	1	1	1	1	0	1	1	1	1
036	0	0	0	0	0	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	1	1	1	1	0	0
037	0	0	0	0	0	1	1	0	1	0	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	0	1	0	0	0	0	0	1	0	1	1
040	0	0	0	0	0	1	1	0	1	0	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	0	1	0	0	0	0	1	1	1	1	1
041	0	0	0	0	0	1	1	0	1	0	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	0	1	0	0	0	1	0	1	1	1	1
042	0	0	0	0	0	1	1	0	1	0	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	0	1	0	0	0	1	1	1	1	1	1
043	0	0	0	0	0	1	1	0	1	0	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	0	1	0	0	1	0	0	1	1	1	1

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Table II

Sector	Tp	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Txs	Tx
	Sector									Sector					T.T								F.D.			Sector											
044	0	0	0	0	0	1	1	0	1	0	0	1	0	1	1	1	1	1	0	0	0	1	0	1	0	0	1	0	0	1	0	1	1	1	1	1
045	0	0	0	0	0	1	1	0	1	0	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1	0	1	0	0	1	1	0	1	1	1	1
046	0	0	0	0	0	1	1	0	1	0	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0	0	1	0	0	1	1	1	1	1	1	1
047	0	0	0	0	0	1	1	0	1	0	1	0	0	0	1	0	0	1	1	0	0	0	1	1	1	0	1	0	1	0	0	0	1	1	1	1
050	0	0	0	0	0	1	1	0	1	0	1	0	0	1	1	1	1	0	0	0	0	0	1	1	0	0	1	0	1	0	0	1	1	1	1	1
051	0	0	0	0	0	1	1	0	1	0	1	0	1	0	0	1	1	1	0	0	1	1	1	0	1	0	1	0	1	0	1	0	1	1	1	1
052	0	0	0	0	0	1	1	0	1	0	1	0	1	1	0	0	1	1	1	0	1	1	1	1	0	0	1	0	1	0	1	1	1	1	1	1
053	0	0	0	0	0	1	1	0	1	0	1	1	0	0	1	1	0	0	0	0	0	0	0	1	0	0	1	0	1	1	0	0	1	1	1	1
054	0	0	0	0	0	1	1	0	1	0	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0	1	0	1	1	0	1	1	1	1	1
055	0	0	0	0	0	1	1	0	1	0	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	0	1	0	1	1	1	0	1	1	1	1
056	0	0	0	0	0	1	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	0	1	1	1	1	1	1	1	1
057	0	0	0	0	0	1	1	0	1	1	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	0	1	1	0	0	0	0	1	1	1	1
060	0	0	0	0	0	1	1	0	1	1	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	0	1	1	0	0	0	1	1	1	1	1
061	0	0	0	0	0	1	1	0	1	1	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	0	1	1	0	0	1	0	1	1	1	1
062	0	0	0	0	0	1	1	0	1	1	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	0	1	1	0	0	1	1	1	1	1	1
063	0	0	0	0	0	1	1	0	1	1	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	0	1	1	0	1	0	0	1	1	1	1
064	0	0	0	0	0	1	1	0	1	1	0	1	0	1	1	1	1	1	0	0	0	1	0	1	0	0	1	1	0	1	0	1	1	1	1	1
065	0	0	0	0	0	1	1	0	1	1	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1	0	1	1	0	1	1	0	1	1	1	1
066	0	0	0	0	0	1	1	0	1	1	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0	0	1	1	0	1	1	1	1	1	1	1
067	0	0	0	0	0	1	1	0	1	1	1	0	0	0	1	0	0	1	1	0	0	0	1	1	1	0	1	1	1	0	0	0	1	1	1	1
070	0	0	0	0	0	1	1	0	1	1	1	0	0	1	1	1	1	0	0	0	0	0	1	1	0	0	1	1	1	0	0	1	1	1	1	1
071	0	0	0	0	0	1	1	0	1	1	1	0	1	0	0	1	1	1	0	0	1	1	1	0	1	0	1	1	1	0	1	0	1	1	1	1
072	0	0	0	0	0	1	1	0	1	1	1	0	1	1	0	0	1	1	1	0	1	1	1	1	0	0	1	1	1	0	1	1	1	1	1	1
073	0	0	0	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	0	0	0	0	0	1	0	0	1	1	1	1	0	0	1	1	1	1
074	0	0	0	0	0	1	1	0	1	1	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	0	1	1	1	1	0	1	1	1	1	1
075	0	0	0	0	0	1	1	0	1	1	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	0	1	1	1	1	1	0	1	1	1	1
076	0	0	0	0	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	0	0
077	0	0	0	0	0	1	1	1	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	1	0	0	0	0	0	0	1	0	1	1
100	0	0	0	0	0	1	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	1	0	0	0	0	0	1	1	1	1	1
101	0	0	0	0	0	1	1	1	0	0	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	1	0	0	0	0	1	0	1	1	1	1
102	0	0	0	0	0	1	1	1	0	0	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	1	0	0	0	0	1	1	1	1	1	1
103	0	0	0	0	0	1	1	1	0	0	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	1	0	0	0	1	0	0	1	1	1	1
104	0	0	0	0	0	1	1	1	0	0	0	1	0	1	1	1	1	1	0	0	0	1	0	1	0	1	0	0	0	1	0	1	1	1	1	1
105	0	0	0	0	0	1	1	1	0	0	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1	1	0	0	0	1	1	0	1	1	1	1
106	0	0	0	0	0	1	1	1	0	0	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0	1	0	0	0	1	1	1	1	1	1	1
107	0	0	0	0	0	1	1	1	0	0	1	0	0	0	1	0	0	1	1	0	0	0	1	1	1	1	0	0	1	0	0	0	1	1	1	1

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Table II

Sector	Tp	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Txs	Tx
	Sector																T.T.						F. D.				Sector										
110	0	0	0	0	0	1	1	1	0	0	1	0	0	1	1	1	1	0	0	0	0	0	1	1	0	1	0	0	1	0	0	1	1	1	1	1
111	0	0	0	0	0	1	1	1	0	0	1	0	1	0	0	1	1	1	0	0	1	1	1	0	1	1	0	0	1	0	1	0	1	1	1	1
112	0	0	0	0	0	1	1	1	0	0	1	0	1	1	0	0	1	1	1	0	1	1	1	1	0	1	0	0	1	0	1	1	1	1	1	1
113	0	0	0	0	0	1	1	1	0	0	1	1	0	0	1	1	0	0	0	0	0	0	0	1	0	1	0	0	1	1	0	0	1	1	1	1
114	0	0	0	0	0	1	1	1	0	0	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	1	0	0	1	1	0	1	1	1	1	1
115	0	0	0	0	0	1	1	1	0	0	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	1	0	0	1	1	1	0	1	1	1	1
116	0	0	0	0	0	1	1	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	1	1	1	1	1	1	1
117	0	0	0	0	0	1	1	1	0	1	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	1	0	1	0	0	0	0	1	1	1	1
120	0	0	0	0	0	1	1	1	0	1	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	1	0	1	0	0	0	1	1	1	1	1
121	0	0	0	0	0	1	1	1	0	1	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	1	0	1	0	0	1	0	1	1	1	1
122	0	0	0	0	0	1	1	1	0	1	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	1	0	1	0	0	1	1	1	1	1	1
123	0	0	0	0	0	1	1	1	0	1	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	1	0	1	0	1	0	0	1	1	1	1
124	0	0	0	0	0	1	1	1	0	1	0	1	0	1	1	1	1	1	0	0	0	1	0	1	0	1	0	1	0	1	0	1	1	1	1	1
125	0	0	0	0	0	1	1	1	0	1	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1	1	0	1	0	1	1	0	1	1	1	1
126	0	0	0	0	0	1	1	1	0	1	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0	1	0	1	0	1	1	1	1	1	1	1
127	0	0	0	0	0	1	1	1	0	1	1	0	0	0	1	0	0	1	1	0	0	0	1	1	1	1	0	1	1	0	0	0	1	1	1	1
130	0	0	0	0	0	1	1	1	0	1	1	0	0	1	1	1	1	0	0	0	0	0	1	1	0	1	0	1	1	0	0	1	1	1	1	1
131	0	0	0	0	0	1	1	1	0	1	1	0	1	0	0	1	1	1	0	0	1	1	1	0	1	1	0	1	1	0	1	0	1	1	1	1
132	0	0	0	0	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	1	1	1	1	0	1	0	1	1	0	1	1	1	1	1	1
133	0	0	0	0	0	1	1	1	0	1	1	1	0	0	1	1	0	0	0	0	0	0	0	1	0	1	0	1	1	1	0	0	1	1	1	1
134	0	0	0	0	0	1	1	1	0	1	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	1	0	1	1	1	0	1	1	1	1	1
135	0	0	0	0	0	1	1	1	0	1	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	1	1
136	0	0	0	0	0	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	0
137	0	0	0	0	0	1	1	1	1	0	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	1	0	0	0	0	0	1	0	1	1	1
140	0	0	0	0	0	1	1	1	1	0	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	1	1	0	0	0	0	1	1	1	1	1
141	0	0	0	0	0	1	1	1	1	0	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	1	1	0	0	0	1	0	1	1	1	1
142	0	0	0	0	0	1	1	1	1	0	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	1	1	0	0	0	1	1	1	1	1	1
143	0	0	0	0	0	1	1	1	1	0	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	1	1	0	0	1	0	0	1	1	1	1
144	0	0	0	0	0	1	1	1	1	0	0	1	0	1	1	1	1	1	0	0	0	1	0	1	0	1	1	0	0	1	0	1	1	1	1	1
145	0	0	0	0	0	1	1	1	1	0	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1	1	1	0	0	1	1	0	1	1	1	1
146	0	0	0	0	0	1	1	1	1	0	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0	1	1	0	0	1	1	1	1	1	1	1
147	0	0	0	0	0	1	1	1	1	0	1	0	0	0	1	0	0	1	1	0	0	0	1	1	1	1	1	0	1	0	0	0	1	1	1	1
150	0	0	0	0	0	1	1	1	1	0	1	0	0	1	1	1	1	0	0	0	0	0	1	1	0	1	1	0	1	0	0	1	1	1	1	1
151	0	0	0	0	0	1	1	1	1	0	1	0	1	0	0	1	1	1	0	0	1	1	1	0	1	1	1	0	1	0	1	0	1	1	1	1
152	0	0	0	0	0	1	1	1	1	0	1	0	1	1	0	0	1	1	1	0	1	1	1	1	0	1	1	0	1	0	1	1	1	1	1	1
153	0	0	0	0	0	1	1	1	1	0	1	1	0	0	1	1	0	0	0	0	0	0	0	1	0	1	1	0	1	1	0	0	1	1	1	1
154	0	0	0	0	0	1	1	1	1	0	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	1	1	0	1	1	0	1	1	1	1	1
155	0	0	0	0	0	1	1	1	1	0	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	1	1	0	1	1	1	0	1	1	1	1

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Table II

Sector	Tp	Sector 32	31	30	29	28	27	26	25	24	23	22	T.T. 21	20	19	18	17	16	15	F.D. 14	13	12	11	10	Sector 9	8	7	6	5	4	3	2	1	0	Txs	Tx
156	0	0	0	0	0	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1
157	0	0	0	0	0	1	1	1	1	1	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1
160	0	0	0	0	0	1	1	1	1	1	0	0	0	1	0	0	0	1	0	0	0	1	1	1	0	1	1	1	0	0	0	1	1	1	1	1
161	0	0	0	0	0	1	1	1	1	1	0	0	1	0	0	0	1	1	0	0	0	1	1	0	1	1	1	1	0	0	1	0	1	1	1	1
162	0	0	0	0	0	1	1	1	1	1	0	0	1	1	0	1	1	1	1	0	0	1	1	0	0	1	1	1	0	0	1	1	1	1	1	1
163	0	0	0	0	0	1	1	1	1	1	0	1	0	0	1	0	1	0	1	0	0	1	0	1	1	1	1	1	0	1	0	0	1	1	1	1
164	0	0	0	0	0	1	1	1	1	1	0	1	0	1	1	1	1	1	0	0	0	1	0	1	0	1	1	1	0	1	0	1	1	1	1	1
165	0	0	0	0	0	1	1	1	1	1	0	1	1	0	0	1	0	1	0	0	0	1	0	0	1	1	1	1	0	1	1	0	1	1	1	1
166	0	0	0	0	0	1	1	1	1	1	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0	1	1	1	0	1	1	1	1	1	1	1
167	0	0	0	0	0	1	1	1	1	1	1	0	0	0	1	0	0	1	1	0	0	0	1	1	1	1	1	1	1	0	0	0	1	1	1	1
170	0	0	0	0	0	1	1	1	1	1	1	0	0	1	1	1	1	0	0	0	0	0	1	1	0	1	1	1	1	0	0	1	1	1	1	1
171	0	0	0	0	0	1	1	1	1	1	1	0	1	0	0	1	1	1	0	0	1	1	1	0	1	1	1	1	1	0	1	0	1	1	1	1
172	0	0	0	0	0	1	1	1	1	1	1	0	1	1	0	0	1	1	1	0	1	1	1	1	0	1	1	1	1	0	1	1	1	1	1	1
173	0	0	0	0	0	1	1	1	1	1	1	1	0	0	1	1	0	0	0	0	0	0	0	1	0	1	1	1	1	1	0	0	1	1	1	1
174	0	0	0	0	0	1	1	1	1	1	1	1	0	1	0	1	0	0	0	1	0	0	0	1	0	1	1	1	1	1	0	1	1	1	1	1
175	0	0	0	0	0	1	1	1	1	1	1	1	1	0	1	1	1	0	1	1	1	1	0	1	1	1	1	1	1	1	1	0	1	1	1	1
176	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0
177	0	0	0	0	0	1	1	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	1	1	1	0	0	0	0	0	0	0	1	0	1	1

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TABLE III

Channels as designated in drawings F10526512 and F10526513	Channel information write to read cycle time measured in clock cycles shall be
A	33.5 ± .15
L	33.5 ± .15
N	34.5 ± .15
X	34.5 ± .15
I	34.5 ± .15
D	70.5 ± .15
R	574.5 ± .15
Rmv	286.5 ± .15
Q	574.5 ± .15
Qmv	286.5 ± .15
Main memory channels (Channels prefixed with an M) and sector	4534.43 ± .15

3.5.4.4 Write head current.- Write head current shall correspond to Figure 3 when changing the state of the write head. The write current when writing continuous "ones" or "zeros" shall be 65 ± 1 ma through the write head. These write current requirements shall apply to all tests which require writing operations.

3.5.4.5 Noise.- The noise shall not exceed a maximum of 10 mv, zero-to-peak, when operating under conditions of Table I, when read from the read head.

3.5.5 Saturation.- The difference in amplitude between the two signals at the same location on the memory disk, for all channels except the clock channel, shall not exceed 10 percent when subjected to the following:

(a) A continuously written cyclic pattern of twelve (12) ones (1) followed by twelve (12) zeros (0) followed by (b).

(b) A continuously written cyclic pattern of eleven (11) ones (1) followed by eleven (11) zeros (0).

(c) Using motor power B and read head load B.

3.5.6 Pulse delay.- The pulse delay at the positive 37.5 mv ± 2 mv crossings of a zero potential to a positive potential change between two patterns, for all channels except the clock channel, shall not exceed .20 clock cycles when subjected to the following:

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(a) A continuously written cyclic pattern of twelve (12) ones (1), followed by twelve (12) zeros (0) followed by (b).

(b) A continuously written cyclic pattern of a one (1), followed by a zero (0).

(c) Using motor power B and read head load B.

The write current is to be determined with the memory motor operating and using a memory write head for a load. The lead between the point of measurement, and the memory write head connectors pins for the start or finish terminals shall have no more than 10 pico farads of stray capacitance.

(1) The write current rise time shall not be greater than .09 clock cycles from 6 to 59 milliamps.

(2) The write current over shoot shall not exceed 73 milliamps and it shall not exceed 67 milliamps for greater than .09 clock cycles.

(3) The write current fall time between 59 and 6 milliamps shall not be greater than .09 clock cycles.

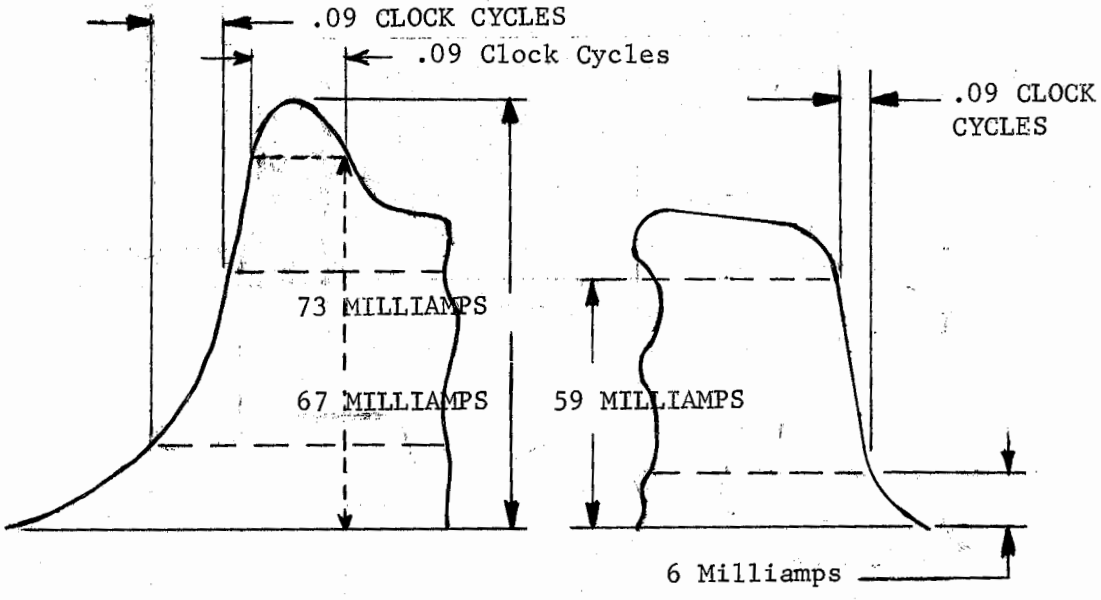


FIGURE 3

WRITE CURRENT WAVEFORM

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3.5.7 Residual magnetism.

3.5.7.1 Amplitude (zero parity bit).- The amplitude at the read head, of the first bit of the word first written, shall not decrease by more than 5 percent under the following conditions:

(a) An alternating pattern of one-zero (1-0) written in a sector (first bit of the word to be a one).

(b) In the sector immediately preceding the sector previously used, a pattern of 10 zeros and a one (00000000001) followed by a zero parity bit.

3.5.7.2 Amplitude (one parity bit).- The amplitude at the read head, of the second bit of the word first written, shall not decrease by more than 5 percent under the following conditions:

(a) An alternating pattern of one-zero (1-0) written in a sector (first bit of the word to be a one).

(b) In the sector immediately preceding the sector previously used, a pattern of 11 zeros (00000000000) followed by a one parity bit.

3.5.8 Power consumption.- When fully assembled, and stabilized at its operating speed, each phase shall be less than 105 watts at a room temperature of 60° to 90°F.

3.6 Memory "run in".- The memory shall be subjected to a 70 hour run in (excluding down time) that will include operation after vibration and operation at temperature extremes. The run in shall be initiated after the unit has satisfactorily met the requirements of 3.2.5, 3.2.6, and 3.5. There shall be a minimum of 125 starts and stops, 50 of which shall be at extreme temperatures. The memory shall continuously meet the requirements of 3.2.5, 3.2.6, and 3.5 during the run in period.

3.6.1 Vibration.- The memory, when mounted in its normal operating position (horizontal) with its mounting pads, and in a 180° rotated position (upside down) with its mounting pads, shall withstand the following vibration frequencies for 30 minutes in each of the above mounting positions. Total time of vibration shall be 1 hour. Vibration frequencies:

(a) Between 0 and 10 cps with an amplitude of 0.030 inch (0.060 inch total excursion).

(b) Between 11 and 20 cps with an amplitude of 0.015 inch (0.030 inch total excursion).

(c) Between 21 and 400 cps with an amplitude constrained to an acceleration of 4.5 gravitational units (g's)

3.6.2 Temperatures.- The memory shall be capable of operation, without damage, for 4 hours when stabilized at an ambient temperature of -22°Fahrenheit (F) and for 4 hours stabilized at 125°F without exceeding a maximum memory temperature of 165°F. The memory shall be in a normal horizontal mounted position for 2 hours of each 4 hour operating period at each temperature, and for the remaining 2 hours the memory shall be in a position rotated 90 degrees from the horizontal. A minimum of 24 starts and stops at each temperature shall be evenly divided between these two positions.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables IV, V, and VI and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.2.4 Functional tests.- The requirements and tests in Table IV shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE IV
(100 percent inspection)

Characteristics	Requirements	Test Procedures
101. Disk rotation	3.2.1	4.5.2.1
102. Residue	3.2.2	4.5.2.2
103. Sealing	3.2.3	4.5.1.6
104. Gassing	3.2.4	4.5.1.7
105. Starting torque	3.2.5	4.5.2.3
106. Speed versus voltage variations	3.2.6	4.5.2.4
107. Memory ground continuity	3.5.1	4.5.2.5
108. Clock channel cycle count	3.5.2.1	4.5.2.6.1

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TABLE IV (Cont'd)

Characteristics	Requirements	Test Procedures
109. Clock channel signal amplitude	3.5.2.2	4.5.2.6.2
110. Clock channel signal frequency (spacing) modulation	3.5.2.3	4.5.2.6.3
111. Sector channel signal amplitude	3.5.3.1	4.5.2.7.1.1, 4.5.2.7.1.2
112. Sector channal, information	3.5.3.2	4.5.2.7.1.2, 4.5.2.7.2
113. Sector channel, noise	3.5.3.3.	4.5.2.8
114. Sector channel, information phase shift	3.5.3.4	4.5.2.9
115. Sector information pulse width	3.5.3.5	4.5.2.10
116. Channel signal amplitude	3.5.4.1	4.5.2.11 (a), (b), (c), (d), (e)
117. Channel pulse width	3.5.4.2	4.5.2.12
118. Channel write head to read head spacing	3.5.4.3	4.5.2.13, 4.5.2.13.1, 4.5.2.13.2
119. Write head current	3.5.4.4	4.5.2.13.3
120. Noise	3.5.4.5	4.5.2.14.1 thru 4.5.2.14.6
121. Saturation	3.5.5	4.5.2.15
122. Pulse delay	3.5.6	4.5.2.16
123. Amplitude (zero parity bit)	3.5.7.1	4.5.2.17.1
124. Amplitude (one parity bit)	3.5.7.2	4.5.2.17.2
125. Power consumption	3.5.8	4.5.2.18
126. Memory "run in"	3.6, 3.2.5, 3.2.6, 3.5	4.5.2.19, 4.5.2.3 thru 4.5.2.18

4.3 Standard sampling.

4.3.1 General.- One memory shall be selected at random by a Government representative as a standard sample from each 100 produced. The sample shall meet the requirements and tests in Table V.

TABLE V

Characteristics	Requirements	Test Procedures
301. Materials	3.1	Certification
302. Design and construction	3.2	Applicable drawings - Visual
303. General specifications	3.3	MIL-M-60185, MIL-C-46357 and MIL-F-13926- Visual

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4.3.2 Environmental.- Three units shall be selected at random by a Government representative as a standard sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall have met the requirements and tests in Tables IV and V and shall then meet the requirements and tests in Table VI.

TABLE VI

Characteristics	Requirements	Test Procedures
304. Vibration	3.4.1	4.5.1.1
305. Shock	3.4.2	4.5.1.2
306. Humidity-Operating	3.4.4.1	4.5.1.3.2
307. Humidity-Non-operating	3.4.4.2	4.5.1.3.1
308. Operating temperature	3.4.3.1, 3.2.5, 3.5	4.5.1.4, 4.5.2.3 thru 4.5.2.1.8
309. Storage temperature	3.4.3.2	4.5.1.4
310. Elevation - operating	3.4.6.1	4.5.2.3 thru 4.5.2.18
311. Elevation - non-operating	3.4.6.2	4.5.2.3 thru 4.5.2.18
312. Vibration	3.6.1	4.5.2.19.1
313. Temperatures	3.6.2	4.5.2.19, 4.5.2.19.2

4.3.3 Failure of sample.- Should any one item of a standard sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the

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design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following sources and equipment shall be utilized in the performance of the applicable tests as specified in 4.5.

Memory power supply.

(a) Motor power A.- 100 volts RMS ± 1.5 percent line to neutral, 3 phase, 400 cycles per second ± 0.5 percent.

(b) Motor power B.- 140 volts RMS ± 1.5 percent line to neutral, 3 phase, 400 cycles per second ± 0.5 percent.

Milliameter.- Range of 0 to 100 ma, accurate to at least 2% of full scale reading.

Oscilloscope.- With a rise time of less than 50 milli-microseconds and an input impedance of 1 meg. ohm or greater and an input capacitance of less than 30 microfarad.

Memory read head load.

(a) Read head load A.- Read head load A shall be equivalent to that shown in Figure 1.

(b) Read head load B.- Read head load B shall be equivalent to that shown in Figure 2.

4.5 Test methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.4.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.4.1 with the memory rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.2.5, 3.2.6 and 3.5.

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4.5.1.2 Shock.- This test shall be conducted in accordance with the conditions of 3.4.2 and Procedure II of the shock test defined in MIL-F-13926 except that impact shocks shall not exceed 5 G's. The memory shall be subsequently inspected for physical damage or defects in compliance with 3.4.2. The unit shall then be inspected as specified in 4.5.1.3 prior to determining compliance with the requirements of 3.2.5, 3.2.6 and 3.5 as required by 3.4.2.

4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.4.3.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.2.5, 3.2.6 and 3.5 as required by 3.4.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.4.3.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 95 ± 5 percent for the 24 hour period, and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.2.3, 4.5.2.4, 4.5.2.5 through 4.5.2.18 to determine compliance with 3.4.4.1.

4.5.1.4 Temperature (operating and storage).- After being subjected to all environmental tests specified above, the memory shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.4.3 shall apply. While at the operating temperatures of 3.4.3.1, the unit shall be tested in accordance with 4.5.2.3, 4.5.2.4, 4.5.2.5 through 4.5.2.18 to determine compliance with the requirements of 3.4.3 as required by 3.4.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2.3, 4.5.2.4, 4.5.2.5 through 4.5.2.18 to determine compliance with 3.4.3.2.

4.5.1.5. Fungus.- To determine compliance with 3.4.5, one of the first 5 produced shall be submitted for testing in accorance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.1.6 Sealing.- The memory shall be subjected to the pressure indicated in 3.2.3 utilizing commercial type testing equipment consisting of pressure measuring gauges, air supply and a test chamber. The chamber containing a memory shall be pressurized to 27 ± 2 inches of mercury gage reading at which time the pressure is shut off. The pressure shall be maintained for the period specified in 3.2.3. Evidence of leakage can be determined by observing the decrease in the gage reading. Any pressure drop in excess of the limit specified for the holding period shall be cause for rejection.

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4.5.1.7 Gassing.- Pressure measuring equipment having a range of at least 32 inches of mercury, a dew point tester and special adapters to accommodate the unit for pressurization shall be utilized. Upon satisfactory completion of the sealing test, the memory shall be flushed in accordance with the requirements of 3.2.4. The flushing can be accomplished by removing one of the four screws number 1052677 shown on the headplate and housing assembly Drawing F10526518. Apply pressure hose with special adapter in the screw hole, and loosen a second of the same numbered screw for the purpose of purging the unit. Flushing shall be continued until the unit has reached a dew point of -25°F or lower as measured with a dew point tester. Lock the screw used for purging, and with the same gas used for flushing, fill the unit until the prescribed pressure as measured on the testing equipment is reached prior to sealing the unit.

4.5.1.8 Elevation.

4.5.1.8.1 Non-operating.- The memory shall be placed within a test chamber and the absolute internal pressure of the chamber reduced at 3.44 inches of mercury (corresponding to an altitude of 50,000 feet above sea level) and the ambient temperature condition specified in 3.4.6.2. The rate of change of pressure shall not exceed 2.44 inches of mercury per minute, and the rate of change of temperature 2 degrees Fahrenheit per minute. The pressure and temperature shall be maintained for the period specified in 3.4.6.2 and shall be returned to the conditions of 3.5 at the same rates. Subsequent to exposure, the memory shall be visually inspected for any evidence of damage, and shall be subjected to the tests defined in 4.5.2.3, 4.5.2.4, 4.5.2.5 through 4.5.2.18 to determine conpliance with 3.4.6.2.

4.5.1.8.2 Operating.- The memory shall be placed within a test chamber and the absolute internal pressure of the chamber reduced to 20.58 inches of mercury (corresponding to an altitude of 10,000 feet above sea level). The rate of change of pressure shall not exceed 2.44 inches of mercury per minute. The temperature specified in 3.4.6.1 shall be maintained. While the altitude and the temperature is maintained, the memory shall be subjected to the tests defined in 4.5.2.3, 4.5.2.4, 4.5.2.5 through 4.5.2.18 to determine compliance with 3.4.6.1.

4.5.2 Performance.- Unless otherwise specified, testing of the memory shall be performed in accordance with the conditions of 3.5.

4.5.2.1 Disk rotation.- Prior to acceptance inspection testing, when a 3 phase, 400 cps ± 0.5 percent 100-140 volts RMS is applied to the unit, it shall be examined visually to determine compliance with 3.2.1.

4.5.2.2 Residue.- With the memory motor power applied, the memory shall be subjected to the operational run in the manner prescribed in 3.2.2. Upon completion of the operational run, the head plates shall be visually examined when the head plate and bracket assembly is removed, to determine compliance with 3.2.2. If iron oxide residue exists on the head plates, the memory shall be cleaned, reassembled and subjected to the complete operational run prescribed in 3.2.2. This procedure shall be repeated until compliance with 3.2.2 is attained. This test shall be accomplished prior to acceptance inspection testing.

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4.5.2.3 Starting torque.- Prior to acceptance inspection testing, starting torque shall be tested using memory motor power A. The memory disk shall achieve a speed in compliance with 3.2.5 within the time period specified therein by means of standard measuring equipment, under Method I and under Method II the specified speed shall be achieved within 20 seconds (see 6.3.7 Methods).

4.5.2.4 Speed versus voltage variations.- After a period of at least one hour of memory motor operation, monitor the sine wave appearing on the clock channel for frequency, and vary the motor voltage to determine compliance with 3.2.6.

4.5.2.5 Ground continuity check.- There shall be electrical continuity less than 1 ohm resistance, between the following points: pin 1 of motor power plug, see drawing D10526509, and both front and rear headplates (determined through any of the head holder hold down screws), pin 1 and the read connector bracket (determined through the connector plug screw), and pin 1 and the front headplate connector bracket (determined through the connector plug screws). The requirements are the same for Method I and II.

4.5.2.6 Clock channel requirements.

4.5.2.6.1 Cycle count.- The positive potential pulses on the clock channel shall be counted in a manner which will demonstrate that there are exactly 4680 negative to positive potential zero crossings per revolution of the disk when tested per Method I or II.

4.5.2.6.2 Signal Amplitude.- The minimum positive or negative pulse amplitude shall not be less than 42.5 millivolts, zero-to-peak per Method I or 45 millivolts, zero-to-peak per Method II. The pulse amplitude shall be determined using read head load A and memory motor power A and the pulse pattern obtained in the cycle count test.

4.5.2.6.3 Signal frequency (spacing) modulation.

4.5.2.6.3.1 Frequency modulation between Q and QR.- To determine compliance with 3.2.5.3, write a cyclic pattern consisting of a one (1), followed by a zero. Write with Q write head and read with QR read head. The pattern shall be written continuously and it shall be demonstrated that the phase shift is less than ± 0.02 clock cycles between a level on the clock wave form which occurs between ± 30 percent of the zero potential, at the leading edge of the positive pulse, and at a level of zero potential ± 27.5 millivolts (mv) being read from the Q channel on the QR read head. The requirements are the same for Method I and II.

4.5.2.6.3.2 Frequency modulation between Q and Qmr.- Repeat test for frequency modulation between Q and QR except read Q channel signal from the Qmr read head. Requirements are the same for Method I and II.

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31

4.5.2.7 Sector channel signal requirements.

4.5.2.7.1 Signal amplitude.

4.5.2.7.1.1 Initial signal amplitude.- The initial sector channel signal amplitude shall be such as to make it possible to meet the requirements of 4.5.2.7.1.2. After the sector channel has met these requirements, the inside radius covered by the sector channel read head shall not be less than 0.003 inches greater than the outside radius covered by the write head lamination of the adjacent inside channel. This check not to be repeated unless read head is moved.

4.5.2.7.1.2 Signal sector (information per Table II) amplitude.- The maximum pulse, zero-to-peak, shall not be greater than 145 millivolts (mv) for Method II and 150 millivolts (mv) for Method I. Amplitude shall be determined using memory motor power B, and read head load B. The minimum pulse, zero-to-peak, shall not be less than 70 mv for Method II and 65.0 mv for Method I. The read head is to be moved to smaller radii to make this adjustment.

4.5.2.7.2 Information.- To determine compliance with 3.5.3.2, the sector channel shall be tested in a manner which will demonstrate that all of the information, per Table II, is contained.

4.5.2.8 Noise.- The sector channel shall be tested in a manner which will demonstrate that there are no pulses, other than those required per Table II, which are greater than 10 millivolts, for Method I, and 9 millivolts for Method II, zero-to-peak and of opposite potential to the last information pulse read. This test shall be performed using motor power B, and read head load B, to determine compliance with 3.5.3.3.

4.5.2.9 Information phase shift.- The sector information phase shift shall be determined on all negative-to-positive potential crossings of the sector information. The phase shift shall be determined at a level of 37.5 millivolts (mv) $\pm$ 2 mv. There shall be .5 $\pm$ .08 clock cycles phase shift with respect to the zero (0) current crossing of a cyclic pattern of one (1) followed by a zero (0), when applied to the sector write head. This test shall be performed using motor power B and read head load B, to determine compliance with 3.5.3.4. These requirements are both for Method I and II.

4.5.2.10 Sector information pulse width.- The pulse width, shall be I $\pm$ 0.12 clock cycles at a channel signal level which is 37.5 millivolts (mv) above zero potential. The pulse width shall be determined using a one (1), which is preceded by four or more zeros (0), and followed by four or more zeros (0). This test shall be performed using motor power A and read head load A, to determine compliance with 3.5.3.5. These requirements are the same for Method I and II.

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4.5.2.11 Other channel signal requirements.

(a) Maximum adjusted signal amplitude.- The read heads shall be moved to a smaller radius to reduce signal amplitude if required to meet amplitude requirements. The maximum pulse amplitude shall not be greater than 150 millivolts for Method I and 145 millivolts for Method II zero-to-peak, reading a cyclic pattern which consists of eight ones followed by eight zeros. The pattern shall be recorded continuously while being read. The amplitude shall be determined using memory motor power B and read head load B.

(b) Minimum adjusted signal amplitude.- The minimum signal amplitude after the maximum signal amplitude has been adjusted (per 4.5.2.11a) shall not be less than 65 millivolts for Method I and 70 millivolts for Method II zero-to-peak. It shall be determined per 4.5.2.11c and 4.5.2.11d.

(c) Minimum adjusted signal amplitude on channels A, L, N, X, I, D, R, Rmr, Q and Qmr.- The minimum adjusted signal amplitude shall be determined while writing continuously a cyclic pattern of one (1) followed by a zero. The signal amplitude shall be determined while using motor power A and read head load A.

(d) Minimum adjusted signal amplitude on all channels that have a number prefixed by an M.- The minimum adjusted signal amplitude shall be determined by continuously writing a cyclic pattern of one (1) followed by a zero. The pattern shall be written in phase in excess of 200 times then written in the opposite phase for one disk revolution. The writing shall be stopped in a manner which shall cause no degeneration of the recorded signal. The signal amplitude shall be determined while using motor power A and read head load A after writing has been stopped.

(e) Average adjusted signal amplitude.- After the average amplitude is set it shall be determined that no channel read head shall cover a radius less than 0.003 inches greater than the outside radius covered by the write head of the adjacent inside channel. This check shall not be repeated unless read heads are moved.

4.5.2.12 Channal pulse width.- The channel pulse width measurement herein are those channels covered by 4.5.2.11. The channel pulse width shall be 1 ± 0.12 clock cycles at a potential which is 37.5±2 millivolts (mv) above zero potential. The pulse width shall be determined while reading a cyclic, one (1) followed by twelve zeros (0), pattern which is being written continuously. The pulse width shall be determined while using motor power A and read head load A for Method I and II.

4.5.2.13 Channel write head to read head spacing.- The period ± 0.2 clock cycles between initiating the writing of a flux transition and the peak of the pulse at the channel read head, produced by that flux transition, shall be equivalent to the period required for the number of clock cycles to be read at the clock read head which is listed for that channel in Table III, in compliance with 3.5.4.3.

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4.5.2.13.1 Course adjustment of head spacing.- Write head to read head spacing is to be determined using motor power A and clock read and channel read head loads A.

4.5.2.13.2 Fine adjustment of write head to read head spacing.- Write continuously a cyclic pattern of one (1) followed by a zero (0). When a negative-to-positive potential change occurs at the read head, the positive 37.5 millivolt (mv) crossing shall lead a zero current crossing at the write head by .5 ± .08 clock cycles for the recirculating loop channels of Table III and by .57 ± .08 clock cycles for the Main Memory Channels of Table III. These requirements shall be met for methods I or II. This shall be determined using motor power A and read head load A.

4.5.2.13.3 Write head current.- Write head current shall correspond to that shown in Figure 3 when changing the state of the write head. The write head current when writing continuous "ones" or "zeros" shall be 65 ± 1 milliamp through the write head. These write current requirements shall apply to all tests which require writing operations, to conform with 3.5.4.4.

4.5.2.14 Adjusted signal channel noise.

4.5.2.14.1 Noise test A.- Write continuously a cyclic pattern of one (1) followed by a zero (0) in phase for more than 100 disk revolutions. Write continuous zeros (0) for one disk revolution. There shall be no pulses in excess of seven millivolts for Method I and 5 millivolts for Method II, zero-to-peak. This shall be determined using motor power B and read head load B.

4.5.2.14.2 Noise test B.- Write continuous zeros (0) for more than 100 disk revolutions. There shall be no pulse, zero-to-peak, greater than 7 millivolts for Method I, and 5 millivolts for Method II. This shall be determined using motor power B and read head load B.

4.5.2.14.3 Noise test C.- Write continuously a cyclic pattern of one (1) followed by seven zeros (0). If the wave form goes positive immediately following the points at which the negative potential pulses go to zero, one of the positive pulses immediately following the negative pulses shall be measured. It shall not be greater than two percent, zero-to-peak, of the preceding negative pulse amplitude, zero-to-peak for both Methods I and II using motor power B and read head load B.

4.5.2.14.4 Noise test D.- Write zeros (0) continuously for 25 disk revolutions or more on the channel being tested, then stop. Read from the channel being tested and write a cyclic pattern of one (1) followed by a zero (0) on the adjacent channel of smaller radii. The signal read from the disk, produced by the adjacent channel write head, shall be less than one millivolts, zero-to-peak, for Method I or II. This is to be measured using motor power B and read head load B.

4.5.2.14.5 Noise test E.- This test applied to channels A, L, N, X, I, D, R, and Q. Read on one channel and write continuously a cyclic pattern of one (1) followed by a zero (0) on all channels of the same radii. There shall be no pulse, zero-to-peak, greater than 7 millivolts for Method I or greater than 5 millivolts for Method II. Motor power B and read head load B shall be used.

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4.5.2.14.6 Total noise.- The sum of the readings determined by noise test A through E, 4.5.2.14.1 through 4.5.2.14.5 for each channel, shall not exceed 10 millivolts (mv) for Methods I or II to determine compliance with 3.5.4.5.

4.5.2.15 Saturation test.- Write continuously a cyclic pattern of twelve (12) ones (1) followed by twelve (12) zeros (0). Write continuously a cyclic pattern of eleven (11) ones (1) followed by eleven (11) zeros (0). The difference in amplitude between the two signals, at the same location on the disk, shall not exceed 10 percent in compliance with 3.5.5.

$$\text{Percent} = \frac{\text{Max.} - \text{Min.}}{\text{Max.}} \times 100$$

This shall be determined by Method II, using motor power B and read head load B.

4.5.2.16 Pulse delay test.- Write continuously a cyclic pattern of twelve (12) ones (1) followed by twelve (12) zeros (0). Write continuously a cyclic pattern of a one (1) followed by a zero (0). The pulse delay at the positive 37.5 millivolt (mv) ± 2 mv crossings, of a zero potential to a positive potential change, between these two patterns, shall not exceed .20 clock cycles in compliance with 3.5.6, for Methods I and II. This shall be determined using motor power B and read head load B.

4.5.2.17 Residual magnetism.

4.5.2.17.1 Amplitude (zero parity bit).-Write an alternating pattern of one-zero (1-0) in a sector (first bit of the word to be a one). Write a pattern of 10 zeros and a one (00000000001) followed by a zero parity bit in the sector immediately preceding the sector previously used. Measure the amplitude at the read head, of the first bit of the word first written to determine compliance with 3.5.7.1.

4.5.2.17.2 Amplitude (one parity bit).- Write an alternating pattern of one-zero (1-0) in a sector (first bit of the word to be a one). Write a pattern of 11 zeros (00000000000) followed by a one parity bit in the sector immediately preceding the sector previously used. Measure the amplitude at the read head, of the second bit of the word first written to determine compliance with 3.5.7.2.

4.5.2.18 Power consumption.- Standard measuring equipment shall be utilized to determine compliance with 3.5.8.

4.5.2.19 Memory "run in".- After each memory is pressure sealed and has successfully passed all performance requirements specified in 3.5.1 through 3.5.8, the memory shall be run for a total of 70 hours excluding down time. The tests shall include at least 4 hours operation at -22°F, 4 hours operation at 125°F and a minimum of 50 hours running time after the -22°F, 125°F and vibration tests have been completed. There shall be a minimum of 125 total starts and stops with at least 25 starts and stops at each extreme temperature.

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4.5.2.19.1 Vibration.- The memory when mounted on its mounting pads shall be vibrated for one (1) hour using the procedure specified in the vibration requirement of 4.5.1.1. One half hour of the vibration shall be accomplished with the memory in the normal operating position with the mounting pads in the horizontal plane, and one half hour shall be accomplished in a position 180° from the normal operating position (upside down). After completing the vibration the memory shall be run for 4 hours and shall meet the requirements specified in 3.5.1 through 3.5.8.

4.5.2.19.2 Temperatures.- The memory shall be placed in a chamber, the temperature of which shall be increased at a rate not to exceed 2° per minute until a temperature of 125°F is reached. A minimum stabilizing period of 4 hours shall be allowed. The memory shall be operated for a minimum operating running time of 4 hours at the 125°F and shall meet all requirements of 3.5.1 through 3.5.8. During this running time a minimum of 25 starts and stops shall be performed while running it at 125°F. Two hours of operation shall be in a normal horizontal position and two hours shall be with the memory rotated 90° from horizontal. A minimum of 12 starts and stops shall be performed in each position.

4.5.2.19.3 Repeat procedure 4.5.2.19.2 for a -22°F operating temperature.

4.5.2.19.4 The remainder of the 70 hour run in shall be run at an ambient temperature between 55°F and 95°F. A minimum of 75 starts and stops shall be performed during this portion of the run in. The total operating running time shall not be less than 50 hours during this portion of the run in, and the memory shall be tested completely for all requirements of 3.5.1 through 3.5.8 immediately before the end of "run in". Every 8 hours during the room temperature "run in" the amplitudes of clock, sector, loops and main memory channels shall meet the requirements specified in 4.5.2.6.1, 4.5.2.7.1, 4.5.2.11. Immediately before the end of the 70 hour "run in" the memory shall be tested for all requirements of 3.5.1 through 3.5.8.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10526600. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The magnetic disk memory has write heads, which are capable of being used to polarize a magnetically hard iron oxide coating on the disk to either of two magnetic states, and read heads which are capable of sensing the magnetic state to which the write heads have polarized the disk coating. The computer uses these capabilities for the storage and retrieval of binary coded information and instructions. The memory also contains a clock track which is used for the synchronizing of all computer operations, and a sector track which is used to determine the angular position of the disk with respect to the sector read head. This makes it possible for the computer to write in specific angular locations and to store an address for this location in a manner which makes it possible to induce the computer to retrieve the stored information at a later time.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Selection of applicable level of preservation, packaging and packing.

(c) Applicable stock number

(d) Applicable packaging data sheet (see 5.1)

(e) A block of serial numbers to identify each magnetic disk

6.3 Definitions.- The definitions which are listed below apply to the following terms when used throughout this document.

6.3.1 Write Heads.- Any head in the magnetic disk memory which is used to alter the magnetic state of the iron oxide coating on disk in the magnetic disk memory.

6.3.2 Read Heads.- Any head which is used to sense flux changes in the iron oxide coating on the disk in the magnetic disk memory.

6.3.3 Channel.- A write head, associated read head and the portion of the oxide coated disk, which is used by these heads.

6.3.4 (One or 1).- The magnetic state of the iron oxide coating on the disk which is produced when positive current flows from center terminal to start terminal of the write head for a period corresponding to one clock cycle. The magnetic flux change which occurs when material polarized to the opposite magnetic state passes under a read head prior to material polarized to the one state shall cause the finish terminal of the read head to be at a positive potential with respect to start terminal.

6.3.5 (Zero or 0).- The magnetic state of the iron oxide coating on the disk which is produced when positive current flows from center terminal to finish terminal of the write head for a period corresponding to one clock cycle. The magnetic flux change, which occurs when material polarized to the opposite magnetic state passes under the read head prior to material which is polarized to the zero state, shall cause the finish terminal of the read head to be at a negative potential with respect to the start terminal.

6.3.6 Clock Cycle.- The term clock cycle as used in this specification is the same as is normally used to describe something which is periodic except that its beginning and ending shall be confined to the portion of the waveform which is between ± 80 percent of the zero to peak amplitude of the waveform in the area which is being used.

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6.3.7 Methods I and II.

(a) Method I - Requirement met at an ambient temperature of -40° and +135°F.

(b) Method II - Requirements met at ambient temperatures between 60° and 90°F; a barometric pressure of 28 to 32 inches of mercury; and a RH up to 90%.

Custodian

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Preparing activity:

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